

Lab Manual: Deploying a Streamlit App on AWS EC2 with Docker

Objective

By the end of this lab, you will:

- Package your Streamlit app into a Docker image.
- Launch and configure an AWS EC2 instance.
- Deploy and run your Streamlit app in a Docker container on EC2.
- Access the app publicly via your EC2 IP.

Prerequisites

- A working **Streamlit app** (`app.py`).
- Python installed locally.
- **Docker** installed locally (to test before deployment).
- An **AWS account** with permissions to launch EC2 instances.
- Basic familiarity with SSH and terminal commands.

Step 1: Prepare Your App

1. Ensure your main file is named: `app.py`
2. Create a `requirements.txt`: `pip freeze > requirements.txt`

This records all dependencies for Docker.

Step 2: Create a Dockerfile

In your project folder, create a file named Dockerfile:

```
# Base image
FROM python:3.9-slim

# Set working directory
WORKDIR /app

# Install dependencies
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

# Copy application files
COPY . .

# Expose Streamlit default port
EXPOSE 8501

# Run Streamlit app
CMD ["streamlit", "run", "app.py", "--server.port=8501", "--server.address=0.0.0.0"]
```

Explanation:

- `python:3.9-slim` → lightweight Python base.
- `EXPOSE 8501` → Streamlit's default port.
- `CMD` → runs the app on `0.0.0.0` so it's accessible externally.

Step 3: Launch an EC2 Instance

1. Log in to **AWS Console** → **EC2** → **Launch Instance**.
2. Choose **Ubuntu 20.04 LTS** (or later).
3. Select **t2.micro** (Free Tier) or higher.
4. Configure **Security Group**:
 - a. Allow inbound traffic on:

- i. Port **22** (SSH)
 - ii. Port **8501** (Streamlit)
5. Launch instance and download the **.pem key file**.

Step 4: Connect to EC2 via SSH

From your local machine:

```
chmod 400 my-key.pem  
ssh -i my-key.pem ubuntu@<EC2_PUBLIC_IP>
```

Step 5: Install Docker on EC2

Once logged into your instance:

```
# Update system packages  
sudo apt update -y  
  
# Install Docker  
sudo apt install -y docker.io  
  
# Enable and start Docker  
sudo systemctl enable docker  
sudo systemctl start docker
```

Docker is now ready on your EC2 instance.

Step 6: Copy Your App to EC2

From your **local machine**, copy files using scp:

```
scp -i my-key.pem -r ./my-streamlit-app  
ubuntu@<EC2_PUBLIC_IP>:/home/ubuntu/
```

Your app folder is now inside EC2 under `/home/ubuntu/my-streamlit-app`.

Step 7: Build and Run Docker Container

On your **EC2 instance**:

```
# Navigate to app folder  
cd my-streamlit-app
```

```
# Build Docker image  
sudo docker build -t streamlit-app .
```

```
# Run container in detached mode with port mapping  
sudo docker run -d -p 8501:8501 streamlit-app
```

Your Streamlit app is now running inside a container.

Step 8: Access the App in Browser

Open: http://<EC2_PUBLIC_IP>:8501 You should see your Streamlit app live.

Verification Checklist

- app.py, requirements.txt, and Dockerfile created.
- EC2 instance launched with security group allowing port **8501**.
- Docker installed and running on EC2.
- App folder copied to EC2.
- Docker image built and container running.
- App accessible via EC2 public IP.

Summary

In this lab, you:

- Packaged your Streamlit app into a **Docker container**.
- Launched and configured an **AWS EC2 instance**.
- Deployed your app inside Docker and exposed it on port 8501.
- Accessed the app publicly via your EC2 IP.

This approach gives you **full control over your environment**, ideal for scaling, adding background tasks, or integrating with other AWS services.